



Cybersecurity

Project 3 Review Questions

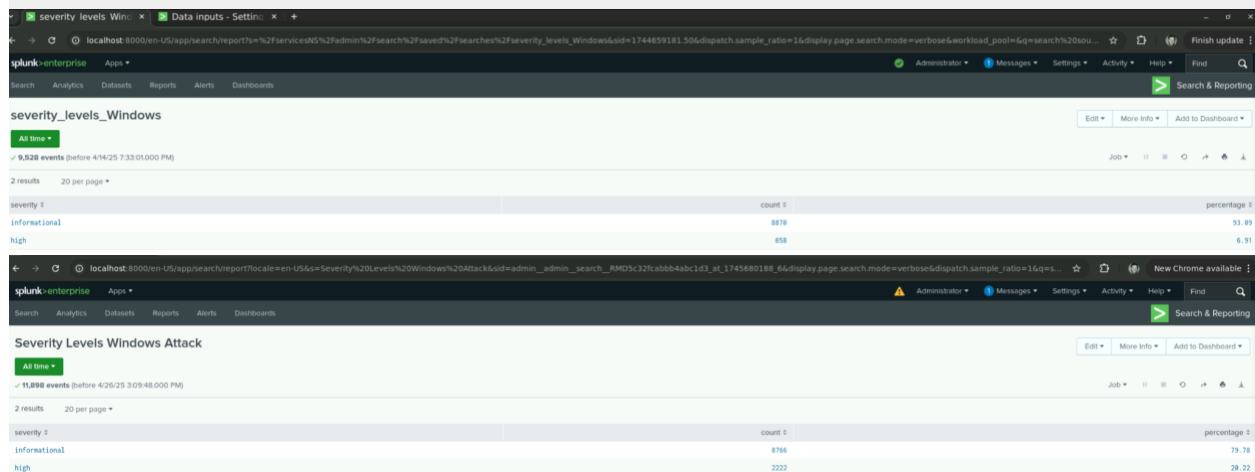
Make a copy of this document before you begin. Place your answers below each question.

Windows Server Log Questions

Report Analysis for Severity

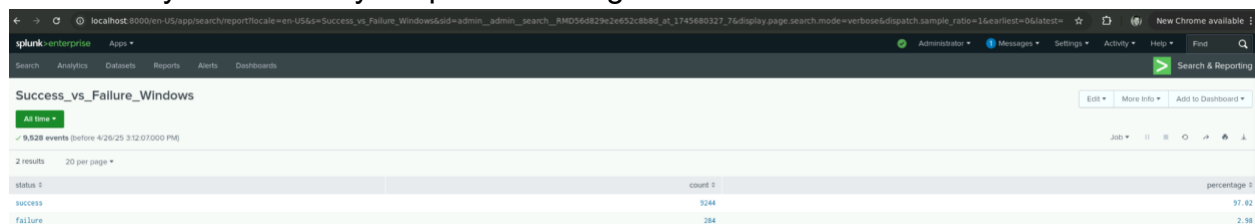
- Did you detect any suspicious changes in severity?

Yes. The levels increased from 7% to 20%.



Report Analysis for Failed Activities

- Did you detect any suspicious changes in failed activities?



Yes

The screenshot shows the Splunk Enterprise search interface. The search bar contains the query: `source="windows_server_attack_logs.csv" host="windows_server_logs" sourcetype="csv" | stats count by status`. Below the search bar, the results are displayed in a table with columns 'status', 'count', and 'percentage'. The table shows two rows: 'success' with a count of 11712 and a percentage of 98.44, and 'failure' with a count of 186 and a percentage of 1.56.

status	count	percentage
success	11712	98.44
failure	186	1.56

Alert Analysis for Failed Windows Activity

- Did you detect a suspicious volume of failed activity?

Yes, the failed activity nearly doubled after the attack.

- If so, what was the count of events in the hour(s) it occurred?

186

- When did it occur?

March 24

- Would your alert be triggered for this activity?

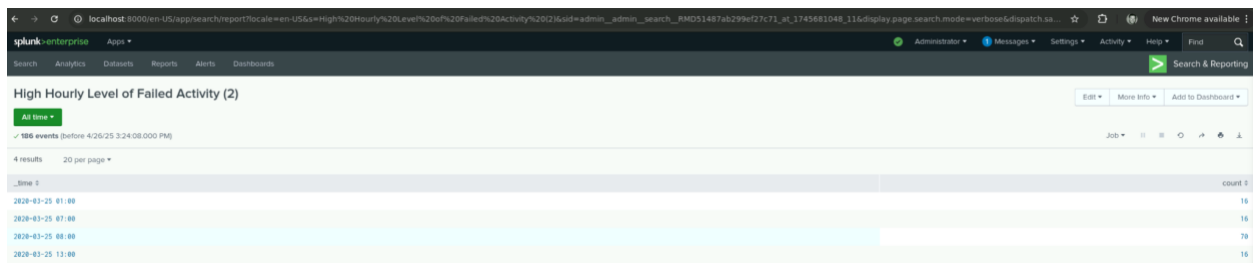
No

- After reviewing, would you change your threshold from what you previously selected?

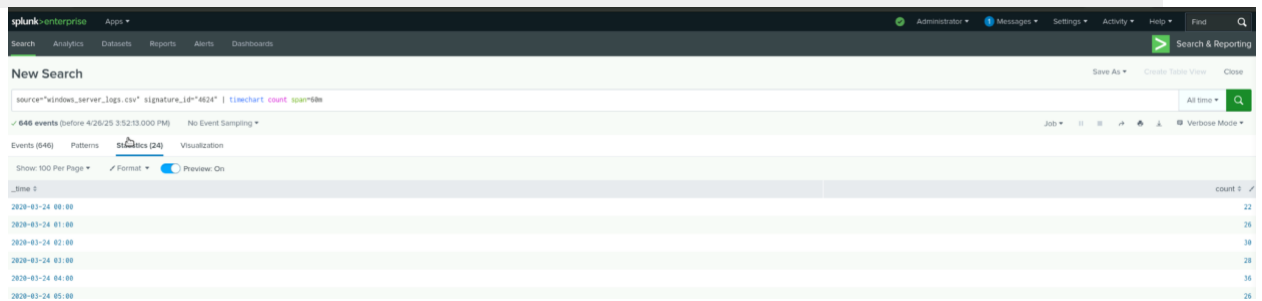
Yes, the alert should be set at 1.

Alert Analysis for Successful Logins

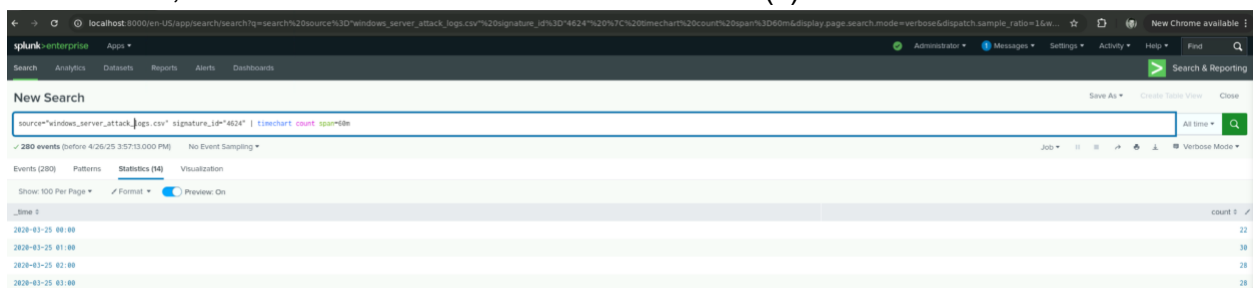
- Did you detect a suspicious volume of successful logins?



Here is the 3/25/20 8am spike



- If so, what was the count of events in the hour(s) it occurred?



Yes, post-attack the logins dropped from 646 to 280.

- Who is the primary user logging in?

User_b and user_1 each had 8 logins

USER

9 Values, 100% of events

Selected Yes No

Reports

Top values

Events with this field

Values	Count	%
user_b	4	18.182%
user_1	4	18.182%
user_d	2	9.091%
user_e	2	9.091%
user_f	2	9.091%
user_h	2	9.091%
user_j	2	9.091%
user_m	2	9.091%
user_n	2	9.091%

- When did it occur?

March 25th, 8am

- Would your alert be triggered for this activity?

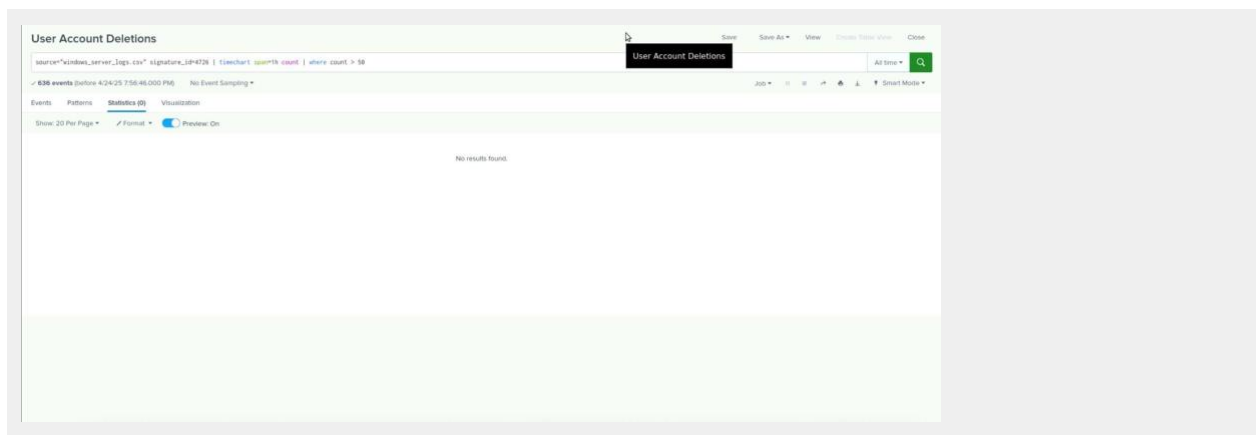
Yes

- After reviewing, would you change your threshold from what you previously selected?

No

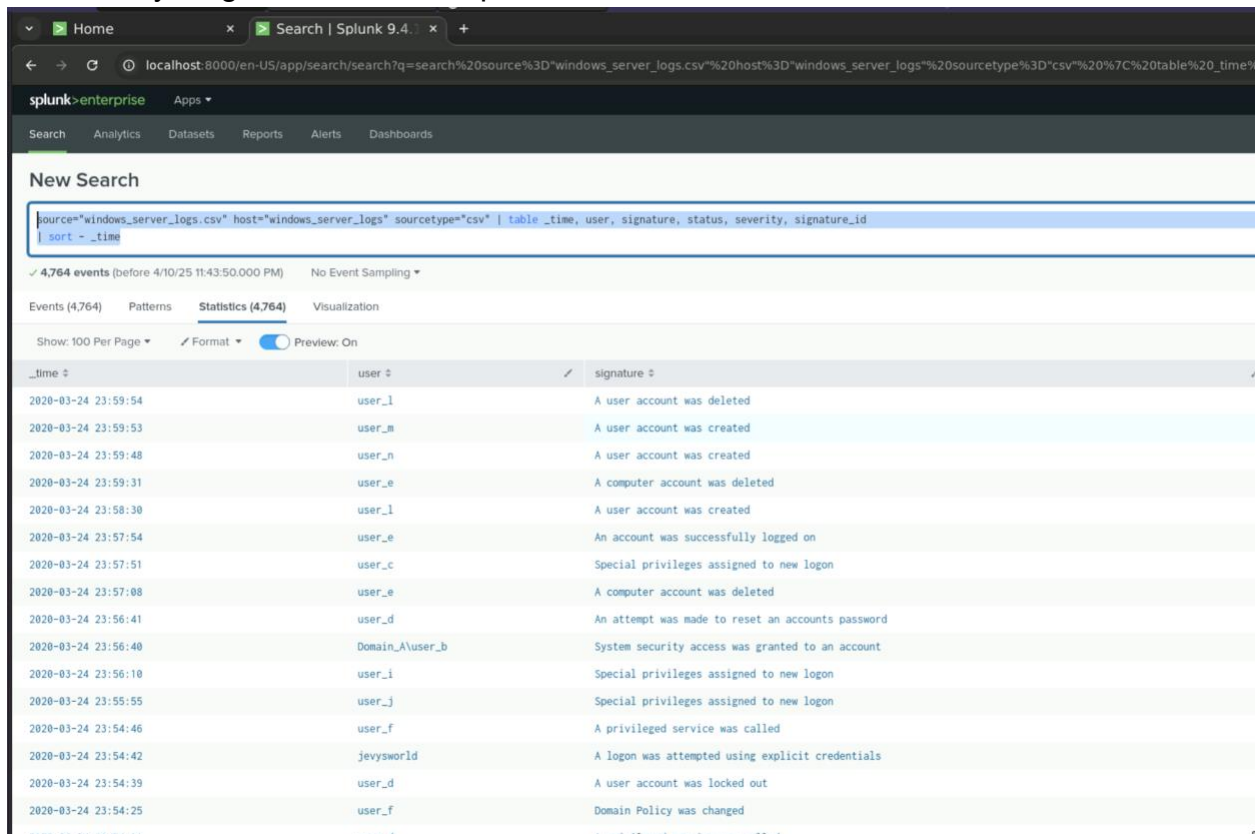
Alert Analysis for Deleted Accounts

Yes. There were 636 deleted accounts in one hour.



Dashboard Analysis for Time Chart of Signatures

- Does anything stand out as suspicious?



The screenshot shows a Splunk search interface with the following search query: `source="windows_server_logs.csv" host="windows_server_logs" sourcetype="csv" | table _time, user, signature, status, severity, signature_id | sort - _time`. The results show 4,764 events. The table below displays a portion of these results, sorted by time in descending order.

_time	user	signature
2020-03-24 23:59:54	user_l	A user account was deleted
2020-03-24 23:59:53	user_m	A user account was created
2020-03-24 23:59:48	user_n	A user account was created
2020-03-24 23:59:31	user_e	A computer account was deleted
2020-03-24 23:58:30	user_l	A user account was created
2020-03-24 23:57:54	user_e	An account was successfully logged on
2020-03-24 23:57:51	user_c	Special privileges assigned to new logon
2020-03-24 23:57:08	user_e	A computer account was deleted
2020-03-24 23:56:41	user_d	An attempt was made to reset an accounts password
2020-03-24 23:56:40	Domain_A\user_b	System security access was granted to an account
2020-03-24 23:56:10	user_i	Special privileges assigned to new logon
2020-03-24 23:55:55	user_j	Special privileges assigned to new logon
2020-03-24 23:54:46	user_f	A privileged service was called
2020-03-24 23:54:42	jevysworld	A logon was attempted using explicit credentials
2020-03-24 23:54:39	user_d	A user account was locked out
2020-03-24 23:54:25	user_f	Domain Policy was changed

Rapid account activity across multiple users and domains

- What signatures stand out?

A user account was changed (4738), created (4720), deleted (4726)

- What is the peak count of the different signatures?

1296, 769 POST, GET

Dashboard Analysis for Users

- Does anything stand out as suspicious?

User Analysis	
User Activity Analysis	
user	Event Count
user_l	708
user_a	564
user_m	550
user_i	542
user_f	540
user_e	538
user_h	538
user_c	534
user_d	528
user_b	526
user_k	520
user_n	516
user_j	512
user_g	478
Domain_A\user_l	188
Domain_A\user_b	186
Domain_A\user_h	184
Domain_A\user_k	180
Domain_A\user_c	96
Domain_A\user_j	96

User_l stood out having the highest (708) event count. User_l's heavy activity was creating and deleting accounts.

The screenshot shows the Splunk Enterprise interface with a search for 'User_L'. The results table displays the following events:

Time	Event
3/24/20 11:55:39.000 AM	2020-03-24T11:55:39.000+0000,"Domain_A\user_l",,"Account Management",,"No - 4024,A user account was successfully logged on,0,.....,Audit Success,8046,.....,Security,....,221802146-1423-3F34-24E7-20C39AC2447),,"8x78E4
3/24/20 11:55:39.000 AM	2020-03-24T11:55:39.000+0000,"Domain_A\user_l",,"Account Management",,"No - 4024,A user account was successfully logged on,0,.....,Audit Success,8046,.....,Security,....,221802146-1423-3F34-24E7-20C39AC2447),,"8x78E4
3/24/20 11:54:38.000 AM	2020-03-24T11:54:38.000+0000,"Domain_A\user_l",,"Account Management",,"No - 4024,A user account was successfully logged on,0,.....,Audit Success,8046,.....,Security,....,221802146-1423-3F34-24E7-20C39AC2447),,"8x78E4
3/24/20 11:54:38.000 AM	2020-03-24T11:54:38.000+0000,"Domain_A\user_l",,"Account Management",,"No - 4024,A user account was successfully logged on,0,.....,Audit Success,8046,.....,Security,....,221802146-1423-3F34-24E7-20C39AC2447),,"8x78E4

- What time did it begin and stop for each user?

The screenshot shows the Splunk Enterprise interface with a search for 'User_L'. The results table displays the following events:

Time	Event
3/24/20 11:59:54.000 PM	2020-03-24T23:59:54.000+0000,"Domain_A\user_l",,"Account Management",,"No - 4024,A user account was successfully logged on,0,.....,Audit Success,8046,.....,Security,....,221802146-1423-3F34-24E7-20C39AC2447),,"8x78E4
3/24/20 11:59:54.000 PM	2020-03-24T23:59:54.000+0000,"Domain_A\user_l",,"Account Management",,"No - 4024,A user account was successfully logged on,0,.....,Audit Success,8046,.....,Security,....,221802146-1423-3F34-24E7-20C39AC2447),,"8x78E4

2020-03-24 23:59:54, user_1

Dashboard Analysis for Signatures with Bar, Graph, and Pie Charts

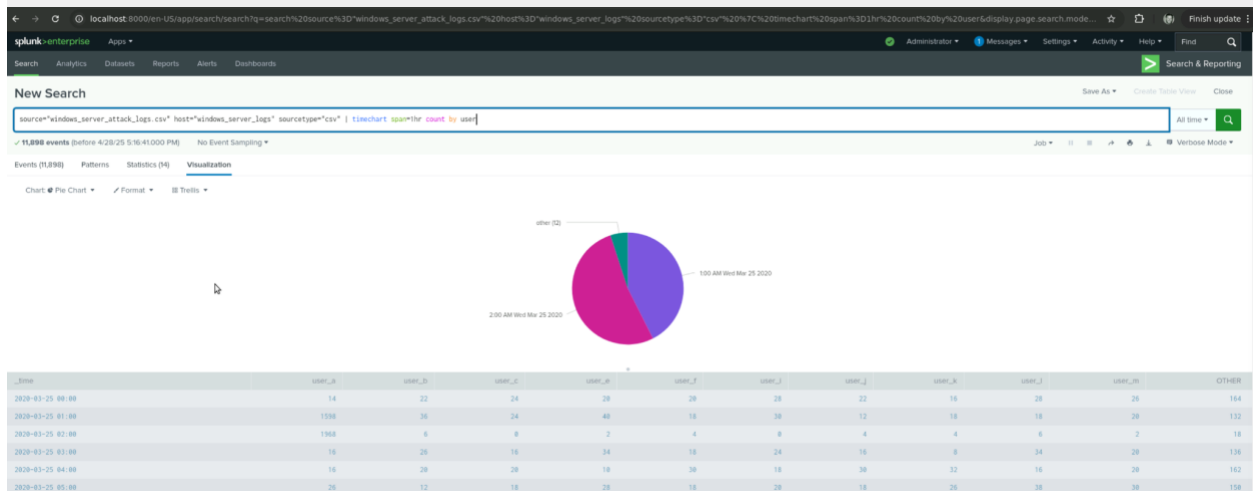
- Do the results match your findings in your time chart for signatures?

Yes

Dashboard Analysis for Users with Bar, Graph, and Pie Charts

- Does anything stand out as suspicious?

User_a stands out as suspicious for heavy activity between 1-2 am on March 25, 2020.



- Do the results match your findings in your time chart for users?

yes

Dashboard Analysis for Users with Statistical Charts

- What are the advantages and disadvantages of using this report, compared to the other user panels that you created?

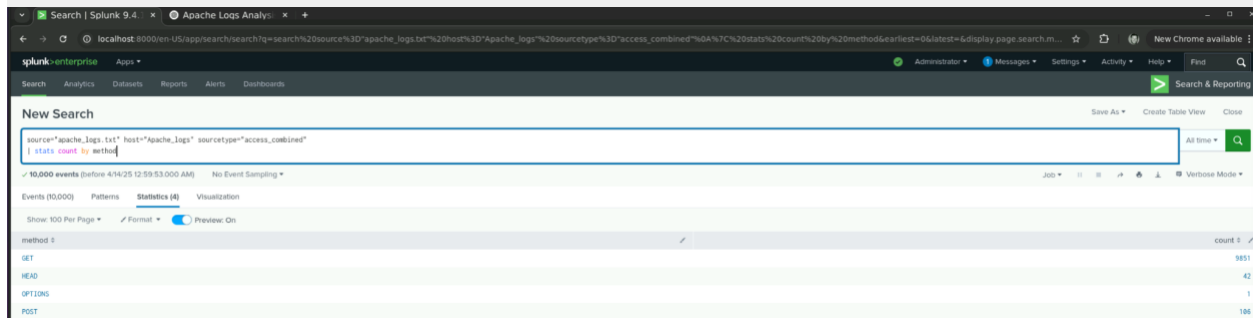
The Splunk dashboard was highly-customizable, allowing for extremely detailed drill-downs and data-heavy operations. A disadvantage for me may have been the statistics background needed for optimal use.

Apache Web Server Log Questions

Report Analysis for Methods

- Did you detect any suspicious changes in HTTP methods? If so, which one?

Yes. The POST method is suspicious because of its relatively low frequency and targeting of a login-related URI. Most of the methods were GET.



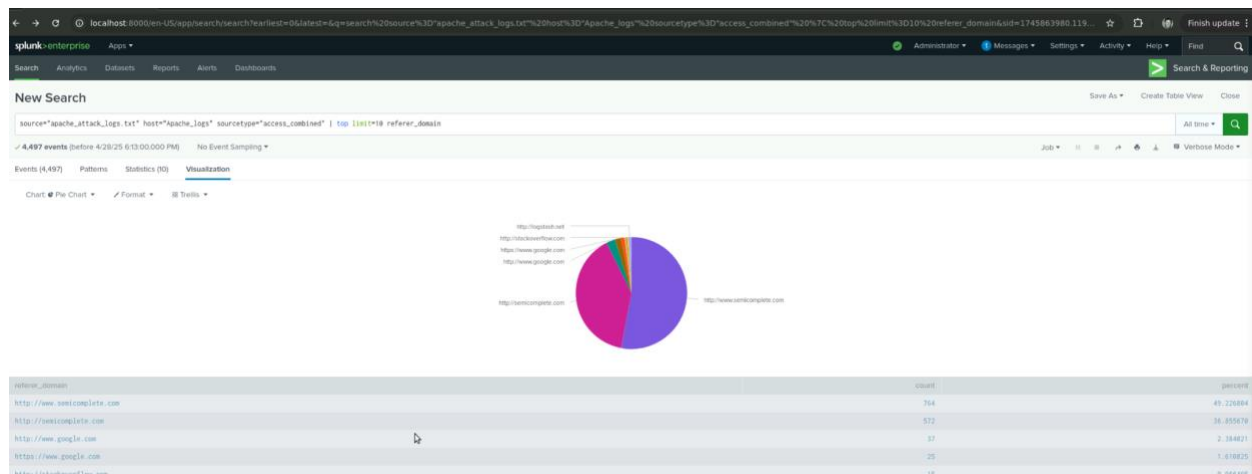
- What is that method used for?

The POST request is used to send data to the server. Authenticating users, uploading forms, etc. It can be used in a brute force attack especially when used on a public-facing login page.

Report Analysis for Referrer Domains

- Did you detect any suspicious changes in referrer domains?

Yes, nearly all traffic was from semicomplete.com



Report Analysis for HTTP Response Codes

- Did you detect any suspicious changes in HTTP response codes?

Yes. The 404 errors may indicate unauthorized access attempts at login. A successful 200 response for the POST to the `VSI_account_logon.php` is also suspicious.

Alert Analysis for International Activity

- Did you detect a suspicious volume of international activity?

The logs show an average of 44.5 requests per day internationally. Geolocation picked up IP addresses from Russia, Romania, Indonesia, Belgium, France, China, and Poland.

- If so, what was the count of the hour(s) it occurred in?

International activity peaked at 10 pm on the night of March 17th.

- Would your alert be triggered for this activity?

No. We set the alert at 50.

- After reviewing, would you change the threshold that you previously selected?

Yes, the spike of 22 logins per hour could indicate the threshold should be around 20 per hour.

Alert Analysis for HTTP POST Activity

- Did you detect any suspicious volume of HTTP POST activity?

There is 1 POST request from March 17th to the VSI's logon page.

- If so, what was the count of the hour(s) it occurred in?

10pm

- When did it occur?

March 17

- After reviewing, would you change the threshold that you previously selected?

Yes, it would be lowered to 1. This instance is also from a suspicious source.

Dashboard Analysis for Time Chart of HTTP Methods

- Does anything stand out as suspicious?

Yes, the POST on the 17th is the only one of its kind and targets the login page.

- Which method seems to be used in the attack?

Login endpoint targeting using the POST method.

- At what times did the attack start and stop?

10.05.46 pm

- What is the peak count of the top method during the attack?

22 GET

Dashboard Analysis for Cluster Map

- Does anything stand out as suspicious?

There was one POST from inside the US

- Which new location (city, country) on the map has a high volume of activity?
(Hint: Zoom in on the map.)

Eastern Europe had a high concentration of activity.

- What is the count of that city?

22 GET requests from Moscow.

Dashboard Analysis for URI Data

- Does anything stand out as suspicious?

A low frequency, successful 200 response on a login is suspicious.

- What URI is hit the most?

The VSI homepage was hit the most frequently. 17 GET requests were made.

- Based on the URI being accessed, what could the attacker potentially be doing?

The attacker could be trying to authenticate credentials via VSI's logon page to gain unauthorized access to VSI's proprietary information.

